

TASK 16: Executor Framework – Task Scheduler

Tools:

- Primary: Java ExecutorService
- Alternatives: ThreadPoolExecutor

Hints / Mini Guide:

1. Replace manual thread creation with ExecutorService.
2. Create fixed thread pool.
3. Submit multiple tasks.
4. Monitor task execution.
5. Gracefully shut down executor.
6. Compare performance with traditional threads.
7. Handle task failures.
8. Demonstrate real-world use.

Deliverables:

- Task scheduling application
- Performance comparison output

Interview Questions Related To Above Task:

- Why Executor framework?
- Types of thread pools?
- Runnable vs Callable?
- What is Future?
- Graceful shutdown?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

